Random Probing for Structural Self-Interactions in Graph Neural Networks

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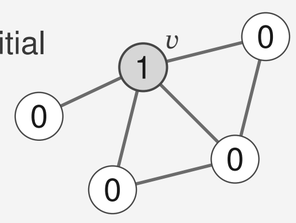
Introduction

- **Limitation:** standard message-passing GNNs cannot express many node-level structural properties
- **DIAL:** a lightweight, modular layer using random features and message passing
- **Expressivity:** closed walks, short cycles, graph angles, neighbor off-diagonals, and polynomial spectral filters
- **Experiments:** accurate cycle counting, strong directed-pattern results, and competitive ZINC performance

Motivation: Cycle Counting

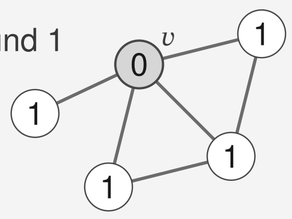
Naive Solution: One-hot Features

Initial



One-hot initial features: $x^{(0)} = e_v$

Round 1

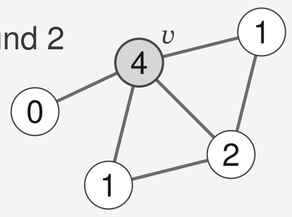


Sum propagation:

$$x_u^{(\ell+1)} = \sum_{w \in N(u)} x_w^{(\ell)},$$

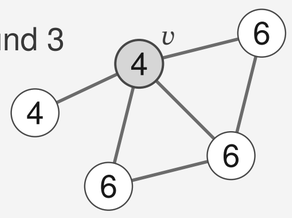
$$x^{(\ell+1)} = Ax^{(\ell)}.$$

Round 2



After k rounds, $x_u^{(k)} = (A^k)_{uv}$ counts length-k walks from v to u.

Round 3



$$x_v^{(3)} = (A^3)_{vv} = 2\,\#\{\text{triangles at } v\}$$

*Can we do message-passing based cycle counting **in parallel for all nodes**?*

Preliminaries

Definition [Rademacher random variables]

$z_1, \dots, z_N \in \{-1, 1\}$ independently, with

$$\mathbb{P}(z_i = -1) = \mathbb{P}(z_i = 1) = 0.5$$

Note: $\mathbb{E}[z_i^2] = 1, \qquad \mathbb{E}[z_i z_j] = 0 \quad (i \neq j)$

Theorem [Hutchinson's diagonal estimator]

Let $z_1, \dots, z_N \in \{-1, 1\}$ be i.i.d. Rademacher random variables. For every $M \in \mathbb{R}^{N \times N}$,

$$\mathbb{E}[z \odot (Mz)] = \mathrm{diag}(M).$$

Definition [Polynomial graph filter]

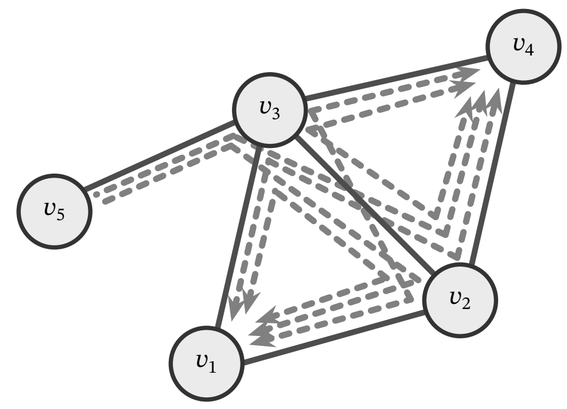
Let S be a graph shift operator. A polynomial graph filter of degree at most $K-1$ is

$$p(S) = \sum_{k=0}^{K-1} \alpha_k S^k.$$

DIAL: Diagonal Access Layer

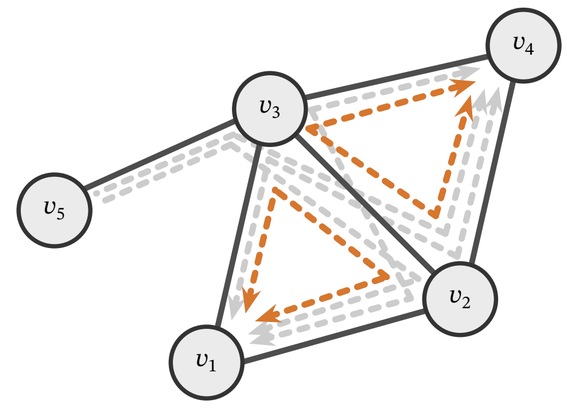
1. Random node probes $p_0 \in \{-1, +1\}^N$

2. Message passing $p_k = A\, p_{k-1}$



Graph shift operator: adjacency or alternatives like Laplacian

3. Correlation with original probe $d_k = p_0 \odot p_k$



Cross-node terms cancel $\mathbb{E}[z_i z_j] = 0 \quad (i \neq j)$

$$\mathbb{E}\big[z_i (A^k z)_i\big] = \sum_{j=1}^{N} (A^k)_{ij}\, \mathbb{E}[z_i z_j]$$

$$= \underbrace{(A^k)_{ii} \mathbb{E}[z_i^2]}_{(A^k)_{ii}} + \underbrace{\sum_{j \neq i} (A^k)_{ij} \mathbb{E}[z_i z_j]}_{0} = (A^k)_{ii}.$$

4. Learn across depths $g = \mathrm{MLP}(d_0, \dots, d_{K-1})$

A shared readout combines information across depths.

5. r probes in parallel $h^{\mathrm{DIAL}} = \mathrm{mean}_{\mathrm{probes}}(g)$

Averaging independent probes reduces variance.

Expressivity

Theorem

DIAL can compute, in expectation,

$$v \longmapsto \big(p(S)\big)_{vv}$$

for every polynomial graph filter p of degree at most $K-1$.

Exact in expectation

- **Closed-walk counts** $\#\,\mathrm{walk}_k(v) = (A^k)_{vv}$
- **Polynomial-filter diagonals** $v \mapsto (p(S))_{vv}$
- **Graph angles** $\angle_\mu(v) = (P_\mu)_{vv}$
- **Neighbor off-diagonal entries** $\{\!\{(p(S))_{vw} : w \in N(v)\}\!\}$
- **Simple cycles of lengths 3, 4, and 5** $C_4(v) = \frac{1}{2}\big[(A^4)_{vv} - d(v)^2 + d(v) - \sum_{w \in N(v)} d(w)\big]$

Approximated by polynomial filters

- **Subgraph centrality** $\mathrm{SC}(v) = (e^A)_{vv}$
- **Heat kernel signature** $\mathrm{HKS}_t(v) = (e^{-tL})_{vv}$

Experiments

- **ZINC logP:** graph-level test MAE
- **Cycle counting:** node-level normalized MAE for 3–6 cycles
- **Directed patterns:** C2–C6, scatter-gather, and bicliques

Probe-based models use $r = 120$ probes and are trained end-to-end.

Node-Level Cycle Counting

Normalized MAE ↓. Mean ± std. over 3 seeds

Cycle	Base GNN	R-PEARL	DIAL
3	0.3515	0.0118 ± 0.0037	$\mathbf{0.0041 \pm 0.0001}$
4	0.2742	0.0360 ± 0.0054	$\mathbf{0.0185 \pm 0.0027}$
5	0.2088	0.0561 ± 0.0035	$\mathbf{0.0271 \pm 0.0005}$
6	0.1555	0.0622 ± 0.0025	$\mathbf{0.0348 \pm 0.0030}$

Higher-order architectures (not shown) are more accurate but also more expensive.

Directed Structural Patterns

Minority-class F1 (%) ↑. Mean ± std. over 5 seeds

Pattern	R-PEARL	DIAL
C2	15.48 ± 7.59	$\mathbf{91.07 \pm 3.33}$
C3	41.21 ± 5.42	$\mathbf{82.44 \pm 4.48}$
C4	66.05 ± 0.52	$\mathbf{79.72 \pm 1.50}$
C5	55.13 ± 0.80	$\mathbf{72.10 \pm 2.21}$
C6	54.74 ± 0.60	$\mathbf{74.83 \pm 2.95}$
S-G	70.54 ± 1.09	$\mathbf{78.68 \pm 2.79}$
B-C	62.56 ± 2.49	$\mathbf{66.07 \pm 1.48}$
Macro F1	52.24	**77.84**

Molecular Prediction on ZINC

Method	Test MAE ↓
B-PEARL	$\mathbf{0.0644 \pm 0.0010}$
DIAL ($K = 4$)	0.0658 ± 0.0020
SPE	0.0693 ± 0.0040
R-PEARL	0.0699 ± 0.0020

Conclusion

- Random probes with **local correlations** reveal node **self-interactions**
- DIAL recovers polynomial-filter diagonals through ordinary message passing
- Expresses closed walks, graph angles, and short-cycle counts in expectation
- Modular, end-to-end layer for standard GNN backbones

References

[1] M. F. Hutchinson. *A stochastic estimator of the trace of the influence matrix for Laplacian smoothing splines.* Communications in Statistics - Simulation and Computation, 19(2):433–450, 1990.

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[3] Charilaos Kanatsoulis, Evelyn Choi, Stefanie Jegelka, Jure Leskovec, and Alejandro Ribeiro. *Learning Efficient Positional Encodings with Graph Neural Networks.* International Conference on Learning Representations (ICLR), 2025.

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Code anonymous.4open.science/r/DIAL **Contact** speltonen@ethz.ch